

Problem 1

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Problem 1 Consider the following limit of Riemann sums of a function g on $[a, b]$:

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n (x_k^* + \cos(x_k^*)) \Delta x, [0, \pi].$$

Express the limit as a definite integral. Use geometry to evaluate the resulting definite integral.